# CHAPTER 43

# THE GAMMA FUNCTION $\Gamma(\nu)$

The *gamma function* is unusual in the simplicity of its recurrence properties. It is because of this that the gamma function (and its special case, the factorial) plays such an important role in the theory of other functions. The reciprocal $1/\Gamma(\nu)$ and the logarithm $\ln\{\Gamma(\nu)\}$ are also important and are discussed in this chapter, as is the related complete beta function $B(\nu,\mu)$, which is addressed in Section 43:13.

Formulas involving the gamma function often become simpler when written for argument $1+\nu$ rather than $\nu$, and we have sometimes taken advantage of this fact. Because $\nu\Gamma(\nu) = \Gamma(1+\nu)$, such a change of argument is readily achieved.

## 43:1 NOTATION

The gamma function is also known as *Euler's integral of the second kind*. $\Gamma(1+\nu)$ is sometimes symbolized $\nu!$ or $\Pi(\nu)$ and termed the *factorial function* or *pi function*, respectively. To avoid possible confusion with the functions of Chapter 45, $\Gamma$ is sometimes distinguished as the *complete gamma function*.

## 43:2 BEHAVIOR

The behavior of $\Gamma(\nu)$ for $-5 < \nu < 6$ is shown on the accompanying map, Figure 43-1; it is complicated. For positive argument, the gamma function passes through a shallow minimum between $\nu = 1$ and $\nu = 2$ and increases steeply as $\nu = 0$ or $\nu = \infty$ is approached. On the negative side, $\Gamma(\nu)$ is segmented: it has positive values for $-2 < \nu < -1$, $-4 < \nu < -3$, $-6 < \nu < -5$, $\cdots$ but negative values for $-1 < \nu < 0$, $-3 < \nu < -2$, $-5 < \nu < -4$, $\cdots$, with discontinuities at $\nu = 0, -1, -2, \cdots$. The gamma function never takes the value zero, but it comes very close between consecutive large negative integers.

The reciprocal $1/\Gamma(\nu)$ has no discontinuities. It rapidly approaches zero as $\nu \to \infty$ and equals zero at $\nu = 0, -1, -2, \cdots$. As shown on the map, its oscillations become increasingly violent as the argument becomes more negative.

The logarithm $\ln\{\Gamma(\nu)\}$ is usually only considered for $\nu > 0$. This is the convention adopted in drawing the map, which shows that $\ln\{\Gamma(\nu)\}$ is a positive function except between $\nu = 1$ and $\nu = 2$, where it is briefly and slightly negative.

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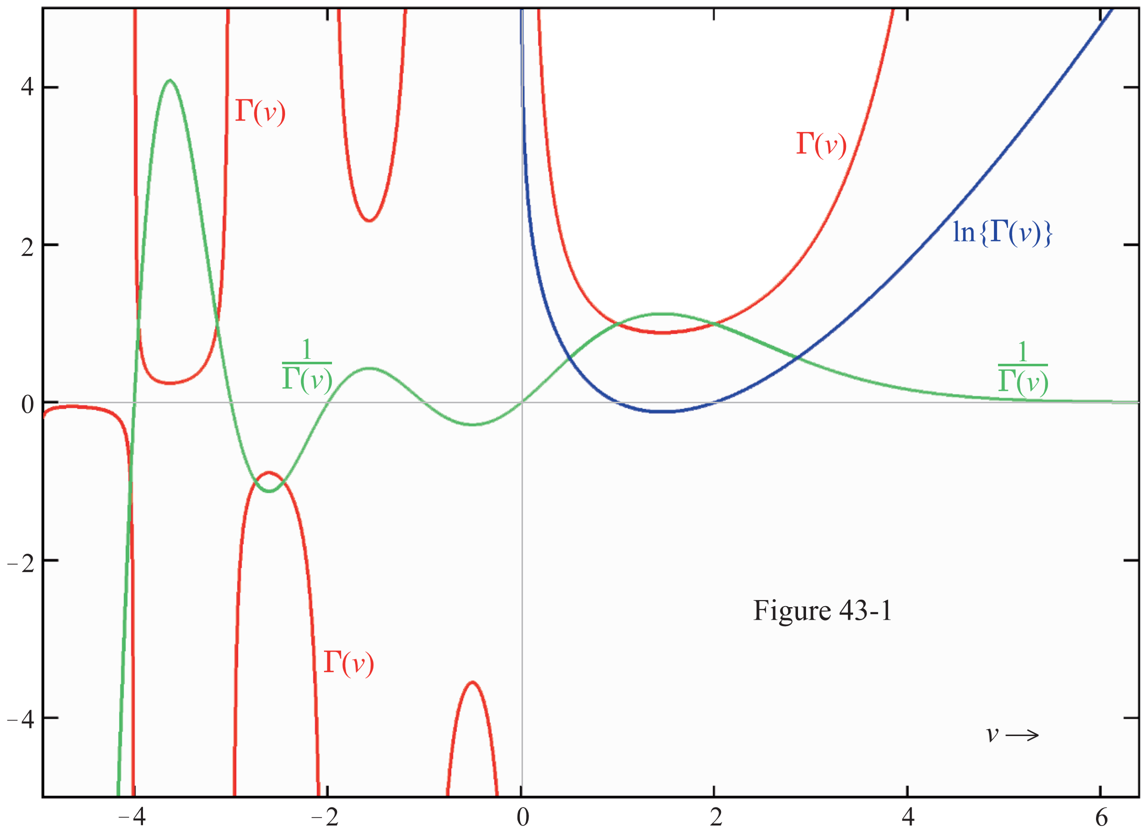


Figure 43-1

## 43:3 DEFINITIONS

Although it is restricted to positive arguments, the most useful definition of the (complete) gamma function is the *Euler integral*

43:3:1
$$\Gamma(v) = \int_0^\infty t^{v-1}\exp(-t)\,\mathrm{d}t \qquad v > 0$$

More comprehensive are the *Gauss limit definition*

43:3:2
$$\Gamma(v) = \lim_{n\to\infty}\left\{\frac{n^v}{v\left(1+v\right)\left(1+\frac{v}{2}\right)\cdots\left(1+\frac{v}{n-1}\right)\left(1+\frac{v}{n}\right)}\right\}$$

and the *infinite product definition of Weierstrass*

43:3:3
$$\frac{1}{\Gamma(v)} = v\exp\left(\gamma v\right)\prod_{j=1}^{\infty}\left(1+\frac{v}{j}\right)\exp\left(\frac{-v}{j}\right)$$

where γ is Euler's constant [Chapter 1].

The (complete) gamma function may be expressed as a definite integral in many ways apart from the one given above. Gradshteyn and Ryzhik [Section 8.31] give a long list of which the following are representative:

43:3:4 $$\Gamma(1+v)=\int_0^1 \ln^v\left(\frac{1}{t}\right)\mathrm{d}t \qquad v>-1$$

43:3:5 $$\Gamma(v)=s^v\int_0^\infty t^{v-1}\exp(-st)\,\mathrm{d}t \qquad v>0 \qquad s>0$$

43:3:6 $$\Gamma(v)=\omega^v\sec\left(\frac{\pi v}{2}\right)\int_0^\infty t^{v-1}\cos(\omega t)\,\mathrm{d}t \qquad 0<v<1 \qquad \omega>0$$

The last two definitions may be regarded as Laplace and Fourier transforms, respectively.

Likewise, there are several ways of representing the logarithm of the gamma function by means of integrals. One is

43:3:7 $$\ln\{\Gamma(1+v)\}=\int_0^1\left(\frac{t^v-1}{t-1}-v\right)\frac{\mathrm{d}t}{\ln(t)} \qquad v>-1$$

and others may be found in Gradshteyn and Ryzhik [Section 8.34].

## 43:4 SPECIAL CASES

The (complete) gamma function reduces to the factorial function [Chapter 2] when its argument is a positive integer:

43:4:1 $$\Gamma(n)=(n-1)! \qquad n=1,2,3,\cdots$$

The (complete) gamma function of odd multiples, positive or negative, of $\frac{1}{2}$ are all proportional to $\sqrt{\pi}$ :

43:4:2 $$\Gamma(\tfrac{1}{2})=\sqrt{\pi}=1.7724\ 53805\ 99052$$

43:4:3 $$\Gamma(n+\tfrac{1}{2})=\frac{(2n-1)!!}{2^n}\sqrt{\pi}=\frac{(2n)!}{4^n\,n!}\sqrt{\pi}=(\tfrac{1}{2})_n\sqrt{\pi} \qquad n=0,1,2,\cdots$$

43:4:4 $$\Gamma(\tfrac{1}{2}-n)=\frac{(-2)^n}{(2n-1)!!}\sqrt{\pi}=\frac{(-4)^n}{(2n)!}\sqrt{\pi}=\frac{(-1)^n}{(\tfrac{1}{2})_n}\sqrt{\pi} \qquad n=0,1,2,\cdots$$

the constant of proportionality being a rational number (an integer or a fraction) involving double factorial [Section 2:13] or factorial [Chapter 2] functions or Pochhammer polynomials [Chapter 18].

In a similar fashion, the (complete) gamma functions of certain multiples of $\frac{1}{3}$ may be expressed in terms of $\Gamma(\frac{1}{3})$ or $\Gamma(\frac{2}{3})$:

43:4:5 $$\Gamma(\tfrac{1}{3})=2.6789\ 38534\ 70775$$

43:4:6 $$\Gamma(\tfrac{2}{3})=\frac{2\pi}{\sqrt{3}\,\Gamma(\tfrac{1}{3})}=1.3541\ 17939\ 42640$$

The proportionality constants in these cases involve triple factorials [Section 2:13] or Pochhammer polynomials [Chapter 18]:

43:4:7 $$\Gamma(n+\tfrac{1}{3})=\frac{(3n-2)!!!\,\Gamma(\tfrac{1}{3})}{3^n}=(\tfrac{1}{3})_n\,\Gamma(\tfrac{1}{3}) \qquad n=0,1,2,\cdots$$

43:4:8 $$\Gamma\left(n+\tfrac{2}{3}\right)=\frac{(3n-1)!!!}{3^n}\Gamma\left(\tfrac{2}{3}\right)=\left(\tfrac{2}{3}\right)_n\Gamma\left(\tfrac{2}{3}\right) \qquad n=0,1,2,\cdots$$

43:4:9 $$\Gamma\left(\tfrac{1}{3}-n\right)=\frac{(-3)^n}{(3n-1)!!!}\Gamma\left(\tfrac{1}{3}\right)=\frac{\Gamma\left(\tfrac{1}{3}\right)}{\left(\tfrac{1}{3}-n\right)_n} \qquad n=0,1,2,\cdots$$

43:4:10 $$\Gamma\left(\tfrac{2}{3}-n\right)=\frac{(-3)^n}{(3n-2)!!!}\Gamma\left(\tfrac{2}{3}\right)=\frac{\Gamma\left(\tfrac{2}{3}\right)}{(\tfrac{2}{3}-n)_n} \qquad n=0,1,2,\cdots$$

The (complete) gamma functions of one quarter and three quarters are related to *Gauss's constant*, $g$ [Section 1:7]:

43:4:11 $$\Gamma\left(\tfrac{1}{4}\right)=\sqrt{(2\pi)^{3/2}\,g}=3.6256\;09908\;22191$$

43:4:12 $$\Gamma\left(\tfrac{3}{4}\right)=\frac{\sqrt{2}\,\pi}{\Gamma\left(\tfrac{1}{4}\right)}=\sqrt{\frac{1}{g}\sqrt{\frac{\pi}{2}}}=1.2254\;16702\;46518$$

Thence, using expressions 43:5:5 or 43:5:6, the (complete) gamma functions of such arguments as $n+\tfrac{1}{4}$ or $\tfrac{3}{4}-n$ may be formulated.

## 43:5 INTRARELATIONSHIPS

The (complete) gamma function obeys the reflection formulas

43:5:1 $$\Gamma(-v)=\frac{-\pi\csc(\pi v)}{v\Gamma(v)}=\frac{\pi\csc(-\pi v)}{\Gamma(1+v)}$$

and

43:5:2 $$\Gamma\left(\tfrac{1}{2}-v\right)=\frac{\pi\sec(\pi v)}{\Gamma(\tfrac{1}{2}+v)}$$

The recurrence formulas

43:5:3 $$\Gamma\left(1+v\right)=v\Gamma(v)$$

and

43:5:4 $$\Gamma\left(v-1\right)=\frac{\Gamma(v)}{v-1}$$

generalize to the formulas

43:5:5 $$\Gamma\left(n+v\right)=v(1+v)(2+v)\cdots(n-1+v)\Gamma(v)=(v)_n\Gamma(v) \qquad n=0,1,2,\cdots$$

and

43:5:6 $$\Gamma\left(v-n\right)=\frac{\Gamma(v)}{(v-1)(v-2)(v-3)\cdots(v-n)}=\frac{(-)^n\Gamma(v)}{(1-v)_n} \qquad n=0,1,2,\cdots$$

involving the Pochhammer polynomial [Chapter 18].

The duplication and triplication formulas

43:5:7 $$\Gamma(2v) = \frac{4^v}{2\sqrt{\pi}}\Gamma(v)\Gamma(\tfrac{1}{2}+v)$$

and

43:5:8 $$\Gamma(3v) = \frac{27^v}{2\pi\sqrt{3}}\Gamma(v)\Gamma(\tfrac{1}{3}+v)\Gamma(\tfrac{2}{3}+v)$$

are the $n = 2$ and 3 cases of the general *Gauss-Legendre formula*

43:5:9 $$\Gamma(nv) = \sqrt{\frac{2\pi}{n}}\frac{n^{nv}}{(2\pi)^{n/2}}\prod_{j=0}^{n-1}\Gamma\left(\frac{j}{n}+v\right) \qquad n = 2,3,4,\cdots$$

which applies for any positive integer multiplier $n$.

From 43:5:5 and 43:5:6, one may derive the expressions

43:5:10 $$\frac{\Gamma(n+v)}{\Gamma(v)} = (v)_n \qquad n = 1,2,3,\cdots$$

and

43:5:11 $$\frac{\Gamma(v-n)}{\Gamma(v)} = \frac{(-1)^n}{(1-v)_n} \qquad n = 1,2,3,\cdots$$

for the ratio of the (complete) gamma functions of two arguments that differ by an integer. These formulas may be used even when the individual gamma functions are infinite; for example, $\Gamma(v-3)/\Gamma(v) \to -\tfrac{1}{6}$ as $v \to 0$.

Because of the frequent occurrence of the reciprocal gamma function in power series expansions of transcendental functions, particular values of the latter functions often serve as sums of infinite series of reciprocal gamma functions. For example, on account of expansion 41:6:4, we have

43:5:12 $$\frac{1}{\Gamma(\tfrac{1}{2})} + \frac{1}{\Gamma(1)} + \frac{1}{\Gamma(\tfrac{3}{2})} + \cdots = \sum_{j=1}^{\infty}\frac{1}{\Gamma(\tfrac{j}{2})} = \frac{1}{\sqrt{\pi}} + e\,\mathrm{erfc}(-1) = 5.5731\ 69664\ 31004$$

while

43:5:13 $$\frac{1}{\Gamma(v)} - \frac{1}{\Gamma(v+1)} + \frac{1}{\Gamma(v+2)} - \cdots = \sum_{j=1}^{\infty}\frac{(-1)^j}{\Gamma(v+j)} = \frac{\gamma\mathrm{n}(v-1,-1)}{e}$$

follows from the expansion in Section 45:5, γn being the *entire incomplete gamma function* [Chapter 45].

## 43:6 EXPANSIONS

Early coefficients in the power series expansion of the (complete) gamma function

43:6:1 $$\Gamma(v) = \sum_{j=1}^{\infty} b_j v^j \qquad v \neq 0,-1,-2,\cdots$$

are $b_1 = 1$, $b_2 = -\gamma$, $b_3 = (6\gamma^2+\pi^2)/12$ and subsequent coefficients may be found from the recursion formula

43:6:2 $$b_{j+1} = \frac{-1}{j}\left[\gamma b_j + (-)^j\sum_{k=1}^{j-1}(-)^k\zeta(j+1-k)b_k\right] \qquad j = 1,2,3,\cdots$$

Here γ denotes Euler's constant [Section 1:7] and $\zeta(n)$ is the $n$th *zeta number* [Chapter 3]. A very similar expansion holds for the reciprocal of the gamma function

43:6:3 $$\frac{1}{\Gamma(v)} = \sum_{j=1}^{\infty} c_j \, v^j$$

and in this case $c_1 = 1$, $c_2 = \gamma$, and $c_3 = (6\gamma^2 - \pi^2)/12$, with a recursion formula,

43:6:4 $$c_{j+1} = \frac{1}{j}\left[\gamma c_j + (-)^j \sum_{k=1}^{j-1} (-)^k \zeta(j+1-k) c_k\right] \qquad j = 1,2,3,\cdots$$

that differs only in a sign from formula 43:6:2. Numerical values of $c_j$ for $j = 1,2,3,\cdots,26$ are listed by Abramowitz and Stegun [page 256].

The power series expansion of the logarithm of the (complete) gamma function is less complicated:

43:6:5 $$\ln\{\Gamma(1+v)\} = -\gamma v + \frac{\pi^2}{12} v^2 - \cdots = -\gamma v + \sum_{j=2}^{\infty} \frac{\zeta(j)}{j} (-v)^j \qquad -1 < v \le 1$$

More rapidly convergent is the similar series

43:6:6 $$\ln\{\Gamma(1+v)\} = (1-\gamma)v + \frac{1}{2}\ln\left(\frac{\pi v(1-v)}{(1+v)\sin(\pi v)}\right) - \sum_{j}^{\infty} \frac{\zeta(j)-1}{j} v^j \qquad j = 3,5,7,\cdots \qquad -1 < v < 1$$

The gamma function may also be expanded as the infinite product

43:6:7 $$\Gamma(1+v) = \prod_{j=1}^{\infty} \frac{(j+1)^v}{(j+v)\, j^{v-1}}$$

An asymptotic expansion of the gamma function is provided by *Stirling's formula* (James Stirling, 1692 – 1770, Scottish mathematician):

43:6:8 $$\Gamma(v) \sim \sqrt{\frac{2\pi}{v}} \exp(-v) v^v \left(1 + \frac{1}{12v} + \frac{1}{288v^2} - \frac{139}{51840v^3} - \cdots\right) \qquad v \to \infty$$

The corresponding asymptotic expansion for the gamma function's logarithm is

43:6:9 $$\ln\{\Gamma(v)\} \sim \ln\left(\sqrt{2\pi}\right) - v + (v - \tfrac{1}{2})\ln(v) + \frac{1}{12v} - \frac{1}{360v^3} + \cdots + \frac{B_{2j}}{2j(2j-1)v^{2j-1}} + \cdots \qquad v \to \infty$$

where $B_{2j}$ denotes a Bernoulli number [Chapter 4]. The series that is part of 43:6:9 is the asymptotic expansion of an integral that defines the *Binet function* (Jacques Phillipe Marie Binet, French physicist and astronomer, 1786 – 1856)

43:6:10 $$\frac{-v}{\pi} \int_0^{\infty} \frac{\ln\left(1 - \exp(-2\pi t)\right)}{v^2 + t^2} \mathrm{d}t \sim \frac{1}{12v} - \frac{1}{360v^3} + \frac{1}{1260v^5} - \cdots = \sum_{j=1} \frac{B_{2j}}{2j(2j-1)v^{2j-1}}$$

Though exact only in the $j \to \infty$ or $v \to \infty$ limit, truncated versions of expansions 43:6:3–9 are remarkably accurate for modest values of the argument [see 43:9:1, for example]. Relationship 43:6:9 is a special case of *Barnes's asymptotic expansion* (Ernest William Barnes, English mathematician and bishop, 1874 – 1953)

43:6:11 $$\ln\{\Gamma(v+c)\} \sim (v+c)\ln(v) - v + \frac{1}{2}\ln\left(\frac{2\pi}{v}\right) + \frac{6c^2 - 6c + 1}{12v} - \frac{2c^3 - 3c^2 + c}{12v^2} + \cdots - \frac{B_{j+1}(c)}{j(j+1)(-v)^j} + \cdots \qquad v \to \infty$$

where $B_{j+1}(c)$ denotes a *Bernoulli polynomial* [Chapter 20]. From this expansion one may also derive the useful expansion

43:6:12 $$\frac{\Gamma(v+c)}{\Gamma(v)} \sim v^c\left[1+\frac{c(c-1)}{2v}+\frac{c(c-1)(c-2)(3c-1)}{24v^2}+\frac{c^2(c-1)^2(c-2)(c-3)}{48v^3}+\cdots\right] \qquad v\to\infty$$

for the ratio of two gamma functions of large, but not very different, arguments.

## 43:7 PARTICULAR VALUES

In addition to those reported in 43:4:5,6 and 43:4:11,12, there are the following:

| $v$ | $-5/2$ | $-2$ | $-3/2$ | $-1$ | $-1/2$ | $0$ | $1/2$ | $1$ | $3/2$ | $2$ | $5/2$ | $3$ | $7/2$ | $\infty$ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| $\Gamma(v)$ | $\frac{-8\sqrt{\pi}}{15}$ | $-\infty\vert+\infty$ | $\frac{4\sqrt{\pi}}{3}$ | $+\infty\vert-\infty$ | $-2\sqrt{\pi}$ | $-\infty\vert+\infty$ | $\sqrt{\pi}$ | $1$ | $\frac{\sqrt{\pi}}{2}$ | $1$ | $\frac{3\sqrt{\pi}}{4}$ | $2$ | $\frac{15\sqrt{\pi}}{8}$ | $\infty$ |

Local maxima or minima of $\Gamma(v)$ correspond to the zeros of the *digamma function* [Section 44:7].

## 43:8 NUMERICAL VALUES

Except at $v=0,-1,-2,\cdots$, (where discontinuities occur) and in the immediate vicinity of these values, *Equator*'s gamma function routine (keyword **Gamma**) returns values of $\Gamma(v)$ precise to 15 decimal digits for inputs in the range $-170 \le v \le 170$. Relation 43:4:1 is exploited if $v$ is a positive integer. Otherwise, *Equator* calculates the integer value $I = \text{Int}(v)$ and the fractional value $f = \text{frac}(v)$ of the input argument, and then employs the algorithm

43:8:1 $$\Gamma(v)\begin{cases}\approx \dfrac{\pi P}{\sin(\pi f)}\displaystyle\sum_{j=1}^{28} c_j(1-f)^j & 0<f<1/2\\ = P\sqrt{\pi} & f=1/2\\ \approx P\Big/\displaystyle\sum_{j=1}^{28} c_j f^j & 1/2<f<1\end{cases}$$

where $c_j$ is the $j$th coefficient in equation 43:6:3. Pochhammer polynomials [Chapter 18] are used as follows

43:8:2 $$P=\begin{cases}1/(v)_{|I|} & I<0\\ 1 & I=0\\ (f)_I & I\ge 1\end{cases}$$

in the calculation of $P$. This strategy is based on equations 43:5:5 and 43:5:1.

*Equator* also has the facility, especially useful when $v$ exceeds 170, to generate values of the logarithm of the gamma function via its ln- or log10-gamma function routines (keywords **lnGamma** and **log10Gamma**). For arguments in the range $0 < v \le 170$, it suffices to take the logarithm of the value generated by the algorithm outlined above. For $v > 170$, *Equator* uses the formula

43:8:3 $$\ln\{\Gamma(v)\} \approx v\ln(v)+\frac{\ln(2\pi/v)}{2}-v+\frac{1}{12v}-\frac{1}{360v^3}+\frac{1}{1260v^5}$$

which is a truncated version of 43:6:9, to produce ample precision. No values of $\ln\{\Gamma(v)\}$ are reported for negative $v$ because $\Gamma(v)$ is itself frequently negative (and hence its logarithm is complex) in this range of argument.

## 43:9 LIMITS AND APPROXIMATIONS

In the limit of large argument,

43:9:1 $$\Gamma(v) \to \sqrt{\frac{2\pi}{v}}\exp\{-v\}v^{v} \qquad v \to \infty$$

this being the leading term in Stirling's approximation 43:6:8.

Close to its zeros, the reciprocal gamma function is approximated by

43:9:2 $$\frac{1}{\Gamma(v)} \approx (-)^{n} n!(v+n)\left[1-(v+n)\psi(n+1)\right] \qquad v \approx -n = 0,-1,-2,\cdots$$

where $\psi$ denotes the digamma function [Chapter 44]. For example, close to the origin, $1/\Gamma(v)$ is well approximated by $v + \gamma v^2$.

The approximation

43:9:3 $$\frac{\Gamma(v+\tfrac{1}{2})}{\Gamma(v)} \approx \sqrt{v}\left[1-\frac{1}{8v}+\frac{1}{128v^{2}}+\frac{5}{1024v^{3}}\right] \qquad \text{large positive } v$$

is sometimes useful because of its rapid convergence. See the discussion surrounding equation 51:13:5 for another aspect of this ratio of gamma functions.

## 43:10 OPERATIONS OF THE CALCULUS

Differentiation of the (complete) gamma function and its logarithm yield

43:10:1 $$\frac{\mathrm{d}}{\mathrm{d}v}\Gamma(v) = \psi(v)\Gamma(v)$$

43:10:2 $$\frac{\mathrm{d}^{n}}{\mathrm{d}v^{n}}\ln\{\Gamma(v)\} = \psi^{(n-1)}(v) \qquad n = 1,2,3,\cdots$$

where $\psi$ and $\psi^{(n)}$ are the digamma and polygamma functions [Chapter 44].

Few simple integrals involving the gamma function have been established, although there are numerous integrals of products and quotients of gamma functions [see Gradshteyn and Ryzhik, Sections 6.41 and 6.42]. A number of integrals involving the logarithm of the gamma function, and including

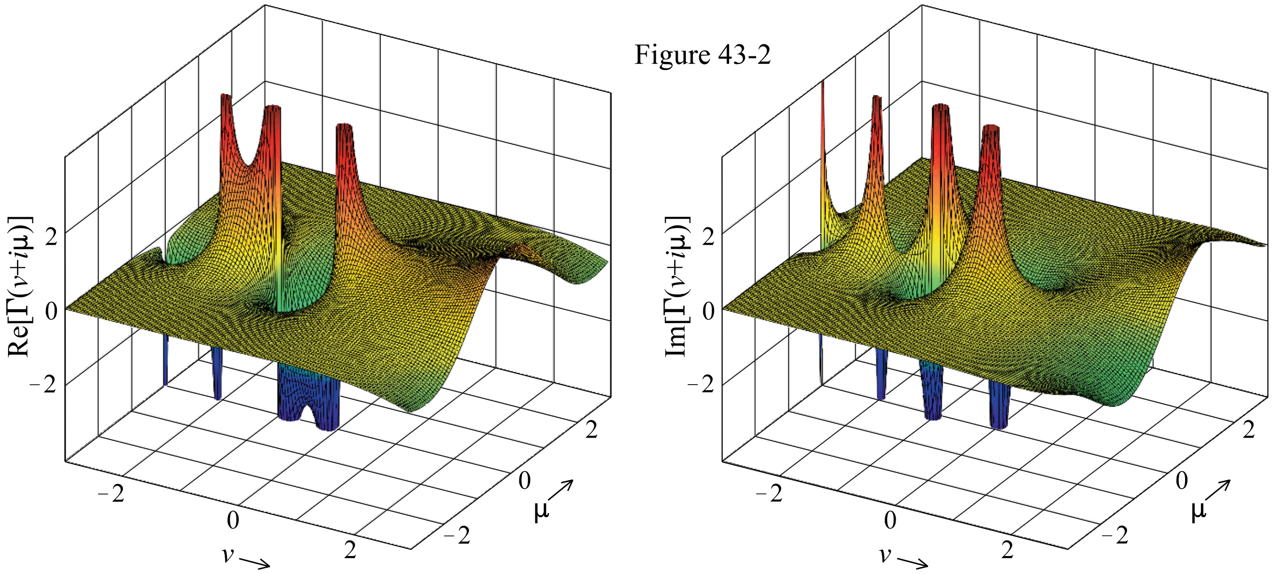
43:10:3 $$\int_{v}^{v+1}\ln\{\Gamma(t)\}\,\mathrm{d}t = \ln\left(\sqrt{2\pi}\right) + v\ln(v) - v \qquad v > 0$$

are also listed by Gradshteyn and Ryzhik [Section 6.44]. The latter formula leads to

43:10:4 $$\int_{1}^{n}\ln\{\Gamma(t)\}\,\mathrm{d}t = \frac{n-1}{2}\left[\ln(2\pi) - n\right] + \sum_{j=2}^{n-1} j\ln(j) \qquad n = 2,3,4,\cdots$$

## 43:11 COMPLEX ARGUMENT

Figure 43-2 portrays the (complete) gamma function of complex variable $v+i\mu$. Note that all the poles of $\Gamma(v+i\mu)$ lie along the nonpositive reaches of the real axis, $\mu = 0$. The real and imaginary components of the complex-valued function are contained in the formulation




43:11:1 $$\Gamma(v+i\mu)=\left[\cos(\theta)+i\sin(\theta)\right]|\,\Gamma(v)\,|\prod_{j=0}^{\infty}\frac{|\,j+v\,|}{\sqrt{\mu^2+(j+v)^2}}$$

where

43:11:2 $$\theta=\mu\,\psi(v)+\sum_{j=0}^{\infty}\frac{\mu}{j+v}-\arctan\left(\frac{\mu}{j+v}\right)$$

Here $\psi$ is the digamma function [Chapter 44]. Tables from which $\Gamma(v+i\mu)$ may be evaluated are given by Abramowitz and Stegun [pages 277−287]. It follows from 43:11:1 that the product $\Gamma(v+i\mu)\Gamma(v-i\mu)$ is real; special cases are

43:11:3 $$\Gamma(1+i\mu)\Gamma(1-i\mu)=\pi\mu\,\mathrm{csch}(\pi\mu)$$

and

43:11:4 $$\Gamma(\tfrac{1}{2}+i\mu)\Gamma(\tfrac{1}{2}-i\mu)=\pi\,\mathrm{sech}(\pi\mu)$$

For purely imaginary argument

43:11:5 $$\Gamma(i\mu)=\sqrt{\frac{\pi}{\mu}\mathrm{csch}(\pi\mu)}\left[\sin(\theta)-i\cos(\theta)\right]\qquad\theta=-\gamma\mu+\sum_{j=1}^{\infty}\frac{\mu}{j}-\arctan\left(\frac{\mu}{j}\right)$$

where $\gamma$ is Euler's constant [Section 1:7].

## 43:12 GENERALIZATIONS

The (complete) gamma function is a special case of the incomplete gamma function $\gamma(v\,,x)$ and of its complement $\Gamma(v,x)$

43:12:1 $$\Gamma(v)=\gamma(v,\infty)\qquad\text{and}\qquad\Gamma(v)=\Gamma(v,0)$$

both of which are discussed in Chapter 45. These two incomplete functions sum to the complete gamma function

43:12:2 $$\gamma(v,x)+\Gamma(v,x)=\Gamma(v)$$

for all values of $x$.

### 43:13 COGNATE FUNCTION: the complete beta function

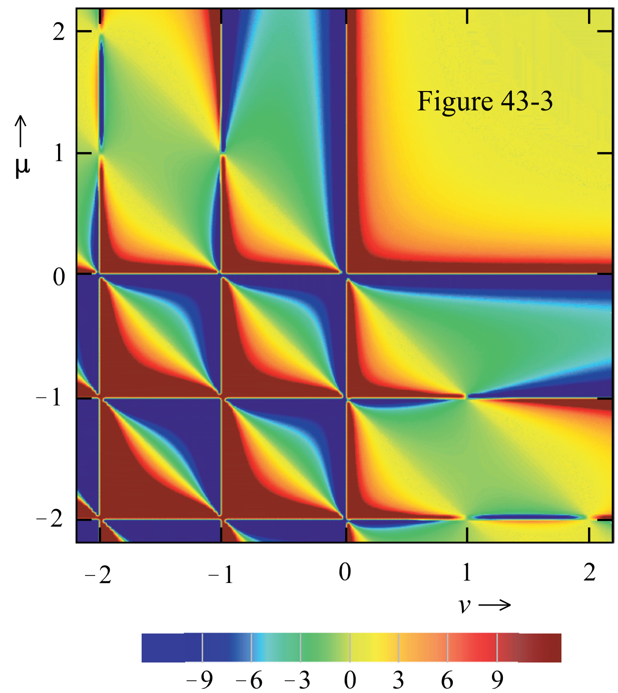
Though it is also related to the functions in Chapters 2, 6, 18, 44 and 45, the closest relative of the (complete) gamma function is the *complete beta function* $\mathrm{B}(\nu,\mu)$, also known as *Euler's integral of the first kind* or simply as the *beta function*. Not to be confused with the beta numbers [Chapter 3], it is defined by the Euler integral

43:13:1 $$\mathrm{B}(\nu,\mu)=\int_0^1 t^{\nu-1}(1-t)^{\mu-1}\mathrm{d}t \qquad \nu>0 \qquad \mu>0$$

and is related to the gamma function through

43:13:2 $$\mathrm{B}(\nu,\mu)=\frac{\Gamma(\nu)\Gamma(\mu)}{\Gamma(\nu+\mu)}$$

The interchangeability $\mathrm{B}(\mu,\nu)=\mathrm{B}(\nu,\mu)$ is implicit in both these definitions and is evident in the symmetry of Figure 43-3. This is a diagram illustrating the "shape" of the function in a region close to the origin.




As with the gamma function, the complete beta function may be expressed as a definite integral in many ways other than 43:13:1. These include

43:13:3 $$\mathrm{B}(\nu,\mu)=\int_0^1\frac{t^{\nu-1}+t^{\mu-1}}{(1+t)^{\nu+\mu}}\mathrm{d}t=\int_1^\infty\frac{t^{\nu-1}+t^{\mu-1}}{(1+t)^{\nu+\mu}}\mathrm{d}t \qquad \nu>0 \quad \mu>0$$

and

43:13:4 $$\mathrm{B}(\nu,\mu)=2\int_0^1 t^{2\nu-1}\left(1-t^2\right)^{\mu-1}\mathrm{d}t=2\int_0^{\pi/2}\sin^{2\nu-1}(t)\cos^{2\mu-1}(t)\,\mathrm{d}t \qquad \nu>0 \qquad \mu>0$$

and extensive lists may be found in Gradshteyn and Ryzhik [Section 8.38] and in Magnus, Oberhettinger and Soni [page 7]. The integral representations of $\mathrm{B}(\nu,\mu)$ apply only when both arguments are positive, but relationship 43:13:2 is valid for any pair of real arguments, as is the infinite-product expression

43:13:5 $$\mathrm{B}(\nu,\mu)=\prod_{j=0}^{\infty}\frac{(j+1)(\nu+\mu+j)}{(\nu+j)(\mu+j)}$$

There are numerous special cases of the beta function that apply when the two arguments have a particular relationship to each other. If $\nu$ and $\mu$ sum to zero, then

43:13:6 $$\mathrm{B}(\nu,-\nu)=\begin{cases}(-1)^\nu/|\nu| & \nu=\pm1,\pm2,\pm3,\cdots\\ \pm\infty & \nu=0\\ 0 & \text{otherwise}\end{cases}$$

or if they sum to a negative integer $-n$, then

43:13:7
$$\mathrm{B}(v,-v-n)=\left\{\begin{array}{ll} 0 & v \text{ not an integer} \\ \dfrac{(-)^{n+v}(-v-n-1)!}{(n+1)_{-n-v}} & v=-n-1,-n-2,-n-3,\cdots \\ \pm\infty & v=0,-1,-2,\cdots,-n \\ \dfrac{(-)^{v}(v-1)!}{(n+1)_{v}} & v=1,2,3,\cdots \end{array}\right\}\quad n=1,2,3,\cdots$$

The uppermost diagonal mostly yellow/green line in Figure 43-3 illustrates the formulas in 47:13:6, while the lines parallel to this are where 47:13:7 hold sway. If one of the arguments is a moiety, if the arguments sum to moiety, or if the two arguments are identical, then the following set of equalities applies to the beta function

43:13:8
$$\frac{4^{v}}{2}\mathrm{B}(v,v)=\cos(\pi v)\mathrm{B}\left(v,\tfrac{1}{2}-v\right)=\mathrm{B}\left(\tfrac{1}{2},v\right)=\sum_{j=0}^{\infty}\frac{(2j-1)!!}{(2j)!!(j+v)}$$

Some important particular values of the beta function are tabulated below

| $\mu$ | $\mathrm{B}(\frac{-5}{2},\mu)$ | $\mathrm{B}(-2,\mu)$ | $\mathrm{B}(\frac{-3}{2},\mu)$ | $\mathrm{B}(-1,\mu)$ | $\mathrm{B}(\frac{-1}{2},\mu)$ | $\mathrm{B}(0,\mu)$ | $\mathrm{B}(\frac{1}{2},\mu)$ | $\mathrm{B}(1,\mu)$ | $\mathrm{B}(\frac{3}{2},\mu)$ | $\mathrm{B}(2,\mu)$ | $\mathrm{B}(\frac{5}{2},\mu)$ |
|---|---|---|---|---|---|---|---|---|---|---|---|
| $5/2$ | $0$ | $\pm\infty$ | $\pi$ | $\pm\infty$ | $-3\pi/2$ | $\pm\infty$ | $3\pi/8$ | $2/15$ | $3\pi/16$ | $4/35$ | $3\pi/128$ |
| $2$ | $4\pi/3$ | $1/2$ | $4\pi/3$ | $\pm\infty$ | $-4$ | $\pm\infty$ | $4/3$ | $1/2$ | $4/15$ | $1/6$ | $4/35$ |
| $3/2$ | $0$ | $\pm\infty$ | $0$ | $\pm\infty$ | $-\pi$ | $\pm\infty$ | $3\pi/4$ | $3/8$ | $\pi/8$ | $4/15$ | $3\pi/16$ |
| $1$ | $-2$ | $-1/2$ | $-2/3$ | $-1$ | $-2$ | $\pm\infty$ | $2$ | $1$ | $3/8$ | $1/2$ | $2/15$ |
| $1/2$ | $0$ | $\pm\infty$ | $0$ | $\pm\infty$ | $0$ | $\pm\infty$ | $\pi$ | $2$ | $3\pi/4$ | $4/3$ | $3\pi/8$ |
| $0$ | $\pm\infty$ | $\pm\infty$ | $\pm\infty$ | $\pm\infty$ | $\pm\infty$ | $\pm\infty$ | $\pm\infty$ | $\pm\infty$ | $\pm\infty$ | $\pm\infty$ | $\pm\infty$ |
| $-1/2$ | $0$ | $\pm\infty$ | $0$ | $\pm\infty$ | $0$ | $\pm\infty$ | $0$ | $-2$ | $-\pi$ | $-4$ | $-3\pi/2$ |

and some others are $\mathrm{B}(\frac{1}{4},\frac{1}{4})=\sqrt{8}\,\pi g$, $\mathrm{B}(\frac{1}{4},\frac{3}{4})=\sqrt{2}\,\pi$ and $\mathrm{B}(\frac{3}{4},\frac{3}{4})=\sqrt{2}/g$, where $g$ is *Gauss's constant* [Section 1:7]. Yet other particular values of the beta function may be found by combining definition 43:13:2 with the findings of Section 43:4. In carrying out this exercise, recognize that the quotient of two gamma functions, both of which are infinite, is generally finite. For example, in view of formula 43:5:6, $\Gamma(-5)/\Gamma(0)=-1/120$.

Though there are exceptions to the rule, it is generally true that the beta function is infinite if one (or both) of its arguments is a nonpositive integer. Likewise, it is generally true that if one (or both) of the arguments is a positive integer $n$, then the beta function reduces to an expression involving a binomial coefficient

43:13:9
$$\mathrm{B}(v,n)=\frac{1}{v\binom{n+v-1}{n-1}}$$

though again, there are exceptions to this rule.

Intrarelationships of complete beta functions, such as

43:13:10
$$\mathrm{B}(v+1,\mu)=\frac{v}{v+\mu}\mathrm{B}(v,\mu)=\frac{v}{\mu}\mathrm{B}(v,\mu+1)$$

as well as expansions, such as

43:13:11 $$\mathrm{B}(\nu,\mu)=\frac{1}{\nu}+\frac{1-\mu}{1+\nu}+\frac{(1-\mu)(2-\mu)}{2(2+\nu)}+\cdots=\sum_{j=0}^{\infty}\frac{(1-\mu)_j}{j!(\nu+j)} \qquad \mu>0$$

and infinite sums, such as

43:13:12 $$\mathrm{B}(\nu,\mu)+\mathrm{B}(\nu+1,\mu)+\mathrm{B}(\nu+2,\mu)+\cdots=\sum_{j=0}^{\infty}\mathrm{B}(\nu+j,\mu)=\mathrm{B}(\nu,\mu-1)$$

may be established via the equation 43:13:2.

*Equator*'s complete beta function routine (keyword **Beta**) first checks for the special cases outlined in equations 43:13:6 and 43:13:7 and sets the value of the function appropriately. If none of those conditions is met, and one or both of the arguments is a nonpositive integer, then *Equator* returns $\pm\infty$. For all other values of the arguments, *Equator* relies on the identity 43:13:2 to generate numerical values of the beta function.

## 43:14 RELATED TOPIC: function synthesis

The (complete) gamma function (or its special case, the factorial) is preeminent as a component of coefficients of power series expansions of the functions of this *Atlas*. The majority of these functions may be regarded as derived from one of four fundamental functions, namely those whose power-series expansions are

43:14:1 $$\sum_{j=0}^{\infty}\frac{x^j}{\Gamma^2(j+1)},\quad \sum_{j=0}^{\infty}\frac{x^j}{\Gamma(j+1)},\quad \sum_{j=0}^{\infty}x^j,\ \text{or}\ \sum_{j=0}\Gamma(j+1)x^j$$

Because $j$ here is a nonnegative integer, the gamma function $\Gamma(j+1)$ is equal to $j!$ or $(1)_j$ and it is the latter Pochhammer notation [Chapter 18] that is most convenient for our present purpose. The four fundamental functions, or prototypes, are:

43:14:2 $$\sum_{j=0}^{\infty}\frac{x^j}{\Gamma^2(j+1)}=\sum_{j=0}^{\infty}\frac{1}{(1)_j(1)_j}x^j=\begin{cases}\mathrm{I}_0\left(2\sqrt{x}\right) & x\ge 0\\ \mathrm{J}_0\left(2\sqrt{-x}\right) & x\le 0\end{cases}$$

43:14:3 $$\sum_{j=0}^{\infty}\frac{x^j}{\Gamma(j+1)}=\sum_{j=0}^{\infty}\frac{1}{(1)_j}x^j=\exp(x)$$

43:14:4 $$\sum_{j=0}^{\infty}x^j=\frac{1}{1-x} \qquad |x|<1$$

and the asymptotic *Euler function*,

43:14:5 $$\sum_{j=0}\Gamma(j+1)x^j=\sum_{j=0}(1)_j x^j\sim\frac{1}{x}\exp\left(\frac{-1}{x}\right)\mathrm{Ei}\left(\frac{1}{x}\right) \qquad x \text{ small}$$

The zero-order Bessel and modified Bessel functions, $\mathrm{J}_0$ and $\mathrm{I}_0$, are treated in Chapters 52 and 49 respectively; the exponential integral function Ei is the subject of Chapter 37.

The four prototype functions specified in equations 43:14:2–5 have been termed *basis hypergeometric functions*, or *prototype hypergeometric functions*. This is because, in the terminology introduced in Section 18:14, they are the simplest members of four families characterized by the difference $L-K$. These latter numbers, $L$ and $K$, are the integers that specify the numbers of Pochhammer polynomials present in the denominator and numerator, respectively, of the hypergeometric representation. Thus, for example, exp($x$), with one denominatorial Pochhammer term and none in its numerator, is the simplest member of the $L = K+1$ family of hypergeometric functions. It is the task of the present section to demonstrate that any two hypergeometric that share the same value of $L-K$ can be interconverted, one to the other. In particular, any hypergeometric function may be formed from its own basis function. For example, any of the functions in Table 18-3 or 18-4 can be formed from the exponential function. This process, which is termed "*synthesis*", was first described in the literature of the fractional calculus [Oldham and Spanier, Chapter 9].

We represent the overall synthetic process by the symbol $\xrightarrow[\beta]{\alpha}$ where α and β are unequal real numbers each of which may be a positive integer or any fractional number, positive or negative. The net effect of this operation is to introduce the Pochhammer polynomial $(\alpha)_j$ into the function's list of numeratorial factors and $(\beta)_j$ into its denominatorial roster; for example:

43:14:6
$$\sum_{j=0}^{\infty}\frac{(a)_j}{(c_1)_j(c_2)_j}x^j \xrightarrow[\beta]{\alpha} \sum_{j=0}^{\infty}\frac{(a)_j(\alpha)_j}{(c_1)_j(c_2)_j(\beta)_j}x^j$$

This is accomplished by three operations applied sequentially to any hypergeometric function:

(i) multiply the input function by $x^{\alpha-1}/\Gamma(\alpha)$;

(ii) apply the differintegration [Section 12:14] operator $\mathrm{d}^{\alpha-\beta}/\mathrm{d}x^{\alpha-\beta}$, and;

(iii) divide by $x^{\beta-1}/\Gamma(\beta)$.

These three steps conspire to achieve the sought goal:

43:14:7
$$\sum_{j=0}^{\infty}\frac{(a)_j x^j}{(c_1)_j(c_2)_j} \xrightarrow{\text{(i)}} \frac{1}{\Gamma(\alpha)}\sum_{j=0}^{\infty}\frac{(a)_j x^{j+\alpha-1}}{(c_1)_j(c_2)_j} \xrightarrow{\text{(ii)}} \frac{1}{\Gamma(\beta)}\sum_{j=0}^{\infty}\frac{(a)_j(\alpha)_j x^{j+\beta-1}}{(c_1)_j(c_2)_j(\beta)_j} \xrightarrow{\text{(iii)}} \sum_{j=0}^{\infty}\frac{(a)_j(\alpha)_j x^j}{(c_1)_j(c_2)_j(\beta)_j}$$

Equation 18:14:18 is the key to understanding step (ii). A straightforward example is the synthesis of Dawson's integral [Chapter 42] from the exponential function.

43:14:8
$$\exp(x)=\sum_{j=0}^{\infty}\frac{1}{(1)_j}x^j \xrightarrow[3/2]{1} \sum_{j=0}^{\infty}\frac{1}{(3/2)_j}x^j = \frac{\exp(x)}{\sqrt{x}}\mathrm{daw}\left(\sqrt{x}\right)$$

Notice that we refer to this process as a synthesis "of Dawson's integral" even though there is a multiplier, here $\exp(x)/\sqrt{x}$, also present and even though the product is $\mathrm{daw}(\sqrt{x})$, not daw($x$).

At first sight, synthesis appears to increase the complement of denominatorial and numeratorial parameters, each by unity, but this is not invariable so because α and/or β may be chosen to match a preexisting parameter, resulting in cancellation. An example is provided by the synthesis

43:14:9
$$\frac{\sin\left(2\sqrt{x}\right)}{2\sqrt{x}}=\sum_{j=0}^{\infty}\frac{1}{(1)_j(3/2)_j}(-x)^j \xrightarrow[3/2]{1} \sum_{j=0}^{\infty}\frac{1}{(3/2)_j(3/2)_j}(-x)^j = \frac{\pi \mathrm{h}_0\left(2\sqrt{x}\right)}{4\sqrt{x}}$$

of a Struve function [Chapter 57]. The difference $L-K$ always remains constant in a synthetic procedure, though the individual $L$ and $K$ integers may increase, remain the same, or even decrease.

It is necessary to ensure that the hypergeometric variable matches that used in the synthetic process, which is $x$ in all our examples throughout the *Atlas*, before invoking synthesis. Therefore function arguments may need adjustment. For example, in synthesizing the cosine function from the zero-order Bessel function, the synthetic operation acts on a function of argument $2\sqrt{x}$

43:14:10
$$\mathrm{J}_0\left(2\sqrt{x}\right) \xrightarrow[1/2]{1} \cos\left(2\sqrt{x}\right)$$

Alternatively, of course, one can synthesize cos($x$) from $\mathrm{J}_0(x)$ by adopting $x^2/4$ as the synthetic variable.

Sometimes more than one synthetic step is needed to create the sought function from the prototypical basis function, as in the example

43:14:11 $$\frac{1}{1-x} \xrightarrow[1]{-1/2} \sqrt{1-x} \xrightarrow[1]{1/2} \frac{2}{\pi}\mathrm{E}\left(\sqrt{x}\right)$$

which leads to a complete elliptic integral of the second kind [Chapter 61]. This example will serve to illustrate that two or more synthetic routes may lead to the same destination:

43:14:12 $$\frac{1}{1-x} \xrightarrow[3/2]{1/2} \frac{\operatorname{artanh}(\sqrt{x})}{\sqrt{x}} \xrightarrow[1]{3/2} \frac{1}{\sqrt{1-x}} \xrightarrow[1]{-1/2} \frac{2}{\pi}\mathrm{E}\left(\sqrt{x}\right)$$

Notice in equations 43:14:2−5 that a limitation sometimes exists on the magnitude of the variable. Thus, in the $L = K$ family of hypergeometric functions, $x$ must have a magnitude of less than unity, and this restriction therefore transfers to the corresponding syntheses, such as those in the two previous equations. The restriction is most telling in the $L = K-1$ family for here the hypergeometric series are asymptotic and the corresponding syntheses, such as that generating the error function complement

43:14:13 $$\frac{-1}{x}\exp\left(\frac{1}{x}\right)\mathrm{Ei}\left(\frac{-1}{x}\right) \xrightarrow[1/2]{1} \sqrt{\frac{\pi}{x}}\exp\left(\frac{1}{x}\right)\mathrm{erfc}\left(\frac{1}{\sqrt{\mathrm{x}}}\right)$$

are valid only in the limit of small $x$.

There exist several differintegration algorithms for zero lower limit [Oldham and Spanier, Chapter 8], though most are restricted to limited ranges of the differintegration order $\mu$. One that is more versatile than most, and without undue complexity, is the *Grünwald algorithm*

43:14:14 $$\frac{\mathrm{d}^{\mu}}{\mathrm{d}x^{\mu}}\mathrm{f}(x) \approx \left(\frac{J}{x}\right)^{\mu}\sum_{j=0}^{J-1}\frac{(-\mu)_j}{j!}\mathrm{f}\left(\frac{J-j}{J}x\right)$$

that uses a large number, $J$, of evenly spaced data points in the range $x/J$ to $x$. Such an algorithm permits differintegration to be carried out numerically. This algorithm may be incorporated into a scheme that implements function synthesis, permitting the synthetic operation

43:14:15 $$\mathrm{f}(x) \xrightarrow[\beta]{\alpha} \mathrm{g}(x)$$

to be carried out numerically, too, as well as algebraically When the three steps specified in 43:14:7 are combined into a single operation, with the differintegration step implemented through the 43:14:14 algorithm, one arrives at

43:14:16 $$\mathrm{g}(x) \approx \frac{\Gamma(\beta)}{J^{\beta-1}\Gamma(\alpha)}\sum_{j=1}^{J}\frac{(\beta-\alpha)_{J-j}}{(J-j)!}\,j^{\alpha-1}\,\mathrm{f}\left(\frac{jx}{J}\right)$$

This last expression is exact in the $J \to \infty$ limit. Though it often delivers a close approximation when $J$ is a sufficiently large positive integer, the formula is unsuitable for calculating precise numerical values of $\mathrm{g}(x)$. Moreover, this algorithm may occasionally fail totally. A case in point is provided in Section 61:3. Whereas synthesis 61:3:6 may be implemented through algorithm 43:14:16, synthesis 61:3:7 cannot be. The failure stems from the lack of differintegrability of an intermediate function.